

# How to load data

# Dealing with PDFs

- PDFs are terrible files for storing data
- Some PDFs are machine-readable, others aren't
  - Machine-readable means the underlying data in the PDF is still intact
  - How can you tell? Try to highlight the text.

# Dealing with PDFs

- If you can highlight the data in the PDF, it **IS** machine-readable
  - Try an online tool like Tabula to get the data into a spreadsheet
- If you can't highlight it, i.e. if it looks like a scanned document or a photo, it's **NOT** machine-readable
  - You'll need OCR (optical character recognition software) to convert into machine-readable data. Try Acrobat or Document Cloud or new Excel hotness
  - OR go back to your data source and ask for a machine-readable file

→ Coming up: Tabula

`tabula.technology`

Tabula is a free, open-source tool that extracts tables from machine-readable PDFs into a spreadsheet.

We'll use it hands-on in a later session — but keep it in mind any time a source hands you a PDF full of data.

# File Types

- Data can be stored in lots of file types
- Text files - the simplest. Basically a text document of data
- File type is usually .txt
  - Obviously a 10,000-page document of data isn't useful. We have to separate it somehow.
  - Generally two ways to do this

# File Types

- Check the file name extension (.csv, .txt, etc.)
- Open the file up in a text editor and browse
- Do you have a header row?
- What's separating each field?

dc311sample.csv

```
X,Y,SERVICEREQUESTID,STREETADDRESS,  
CITY,STATE,ZIPCODE,SERVICECODE,  
SERVICECODEDESCRIPTION,...,  
ADDDATE,RESOLUTIONDATE,...,  
WARD,LATITUDE,LONGITUDE,...
```

```
26-00183801,5224 AMES ST NE,  
WASHINGTON,DC,20019  
2026/03/25 04:00:18+00,  
2026/03/30 14:39:05+00,  
Ward 7
```

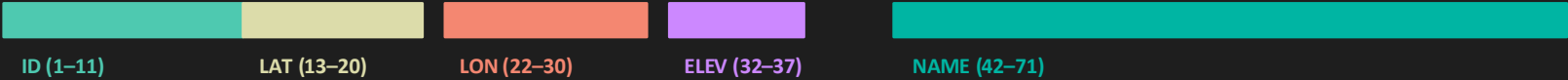
# File Type: Fixed Width

- You must manually put breaks in the data to tell your computer where a new field ends and another begins
- If you have a fixed width file, make sure your source gives you a guide or schema that tells you where to put the breaks. Don't guess!
  - Example

ghcnd-stations.txt - ncei.noaa.gov/pub/data/ghcn/daily/ghcnd-stations.txt

1 2 3 4 5 6 7 8  
12345678901234567890123456789012345678901234567890123456789012345

|             |         |          |        |                       |     |       |
|-------------|---------|----------|--------|-----------------------|-----|-------|
| ACW00011604 | 17.1167 | -61.7833 | 10.1   | ST JOHNS COOLIDGE FLD |     |       |
| AE000041196 | 25.3330 | 55.5170  | 34.0   | SHARJAH INTER. AIRP   | GSN | 41196 |
| AFM00040948 | 34.5660 | 69.2120  | 1791.3 | KABUL INTL            |     | 40948 |
| AG000060390 | 36.7167 | 3.2500   | 24.0   | ALGER-DAR EL BEIDA    | GSN | 60390 |
| ALM00013615 | 41.4150 | 19.7210  | 38.4   | TIRANA RINAS          | GSN | 13615 |



No header row. No delimiters. You must know the column positions from the schema.

# File Type: Delimited

- The data fields are separated by a delimiter (often punctuation) that tells our computer where a new field ends and another begins
  - This is more common than fixed width
  - Common delimiters
    - Commas
    - Tabs
    - Pipes (|)



# File Type: CSV

- Commas are the most common delimiters
- Delimited files that use commas are also known as CSVs, aka comma-separated values
- File type can be .csv instead of .txt
- **But what if you have commas in a field?**
  - A text qualifier (usually quote marks) tells the computer to ignore commas within fields
  - Example: "Trash Cart Repair, urgent" will look like "Trash Cart Repair, urgent" in the text file

```
dc311sample.csv (in text editor)
```

```
SERVICEREQUESTID,STREETADDRESS,SERVICECODEDESCRIPTION
```

```
26-00183801,5224 AMES ST NE,  
Supercan - Delivery
```

```
26-00183803,4528 28TH STREET NW,  
"Trash Cart Repair,  
urgent"
```

```
↑ comma delimiter  
↑ text qualifier (quotes)
```

dc311sample.csv – opened in text editor

```
SERVICEREQUESTID,STREETADDRESS,CITY,STATE,ZIPCODE,SERVICECODE,SERVICECODEDESCRIPTION,...,ADDDATE,RESOLUTIONDATE,WARD
```

```
26-00183801,5224 AMES STREET NE,WASHINGTON,DC,20019,S0422,Supercan - Delivery,...,2026/03/25 04:00:18+00,2026/03/30  
14:39:05+00,Ward 7
```

```
26-00183802,43 ANACOSTIA ROAD NE,WASHINGTON,DC,20019,S0336,Out of State Parking Violation (ROSA),...,2026/03/25  
04:01:35+00,2026/03/25 06:20:07+00,Ward 7
```

```
26-00183803,4528 28TH STREET NW,WASHINGTON,DC,20008,TRACO001,Trash Cart Repair,...,2026/03/25 04:16:01+00,2026/04/07  
20:12:02+00,Ward 3
```

```
26-00183804,1625 ECKINGTON PL NE,WASHINGTON,DC,20002,S0311,Rodent Inspection and Treatment,...,2026/03/25  
04:16:01+00,2026/03/30 14:17:50+00,Ward 5
```

↑ header row

← comma delimiter separates each field

# data FIELD types

Just like there are different types of data files, there are different types of data fields (aka columns aka variables).

How you upload them can make a huge difference.

# data FIELD types

## Text / character

- Alphanumeric
- Short or long
- You don't need to do calculations with these things
- Ex in 311: SERVICEREQUESTID, STREETADDRESS, SERVICECODEDESCRIPTION, WARD

## Numeric

- Things you want to calculate with
- Ex: counts, sums, dollar amounts
- Can be integers or decimal numbers
- Ex in 311: SERVICECALLCOUNT, LATITUDE, LONGITUDE

## Dates / times

- Ex: 2026/03/25 04:00:18+00
- Storing these as dates instead of text lets you calculate days between open and close
- Ex in 311: ADDDATE, RESOLUTIONDATE, SERVICEDUE DATE

# Why should you care about types of data fields?

- **Different field types sort differently!!**
- If you're not careful when you load your data, Excel/Google Sheets will try and be helpful and guess your data types.
  - It's usually wrong.
  - **THUS → DO NOT JUST DOUBLE CLICK YOUR DATA FILE TO OPEN IT!!!!**

# Computers are dumb

How does Excel/Google Sheets get things wrong when guessing field types?

- **Drops leading zeros**
  - DC zip codes like 20001, 20036 are fine — but some jurisdictions use codes starting with 0
  - Excel will assume ZIPCODE is a number and strip leading zeros
  - Avoid this by explicitly loading as a text field
- **Reformat numbers to dates**
  - 311 dates look like: 2026/03/25 04:00:18+00
  - Excel may misread the format and scramble these — or treat them as text strings you can't calculate with
  - Avoid this by explicitly loading as a date field

## Watch out in 311 data

**ZIPCODE** — load as text, not number

**SERVICEREQUESTID** (e.g. 26-00183801) — text

**WARD** — load as text (don't average wards!)

**ADDDATE / RESOLUTIONDATE**  
— explicitly load as dates so you can calculate days to resolution